

Project Design Phase

Solution Architecture

Date	30 June 2026
Team ID	[Team ID]
Project Name	Customer Care Registry
Maximum Marks	4 Marks

Solution Architecture

Solution architecture bridges the gap between the business problem (fragmented, slow customer complaint handling) and the technology solution (the Customer Care Registry platform). It defines the structure, behavior, layers, and components of the system, and how they interact to deliver the required features.

- Find the best tech solution to solve the existing business problem of fragmented, slow, and untracked customer complaint handling.
- Describe the structure, characteristics, and behavior of the Customer Care Registry for all stakeholders (customers, agents, managers, admins).
- Define the layers, components, and data flow that make up the solution.
- Provide a specification against which the solution is built, managed, and delivered.

Solution Architecture Diagram — Customer Care Registry

The diagram below shows the four architectural layers of the Customer Care Registry: the actors who use the system, the React presentation layer, the Express business-logic/API layer (with its functional modules), the AI services layer backed by the Google Gemini API, and the underlying data layer.

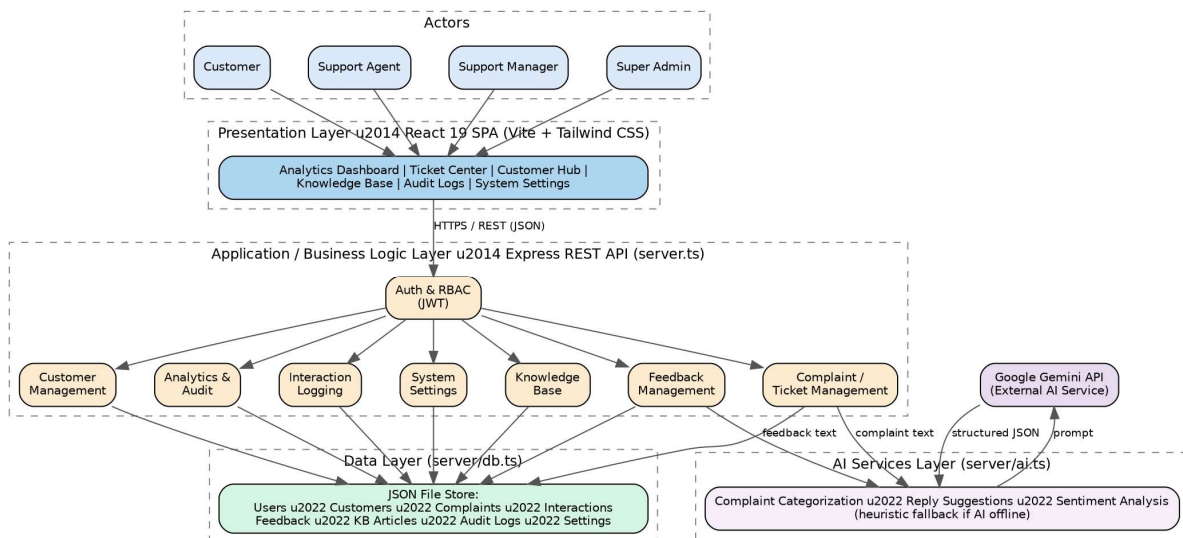


Figure 1: Solution architecture and data flow of the Customer Care Registry

Layer Description

- **Actors:** Customer, Support Agent, Support Manager, and Super Admin — each with a distinct set of permissions enforced by role-based access control.
- **Presentation Layer:** A React 19 + TypeScript single-page application (built with Vite, styled with Tailwind CSS) exposing role-specific modules — Analytics Dashboard, Ticket Center, Customer Hub, Knowledge Base, Audit Logs, and System Settings.

- Application / Business Logic Layer: An Express REST API (server.ts) that authenticates requests via JWT and routes them to the relevant module — customer management, complaint/ticket management, interaction logging, feedback management, knowledge base, analytics & audit, and system settings.
- AI Services Layer: A dedicated AI module (server/ai.ts) that sends complaint and feedback text to the Google Gemini API to auto-suggest category, priority, and department, draft agent replies, and derive feedback sentiment — falling back to keyword heuristics if the AI service is unavailable.
- Data Layer: A JSON file-based data store (server/db.ts) holding users, customers, complaints, interactions, feedback, knowledge base articles, audit logs, and settings — designed so it can be swapped for a relational or NoSQL database without changing the API contract.